RoadInSight: Road Surface Detection Using Inexpensive Cell Phone Sensors

Project Team

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Chapter 1

Introduction

Potholes, bumps and road roughness are among the most annoying hazards and anomalies experienced by commuters and vehicles. Vehicles get damaged in terms of suspension, steering misalignment, tyre punctures etc., which lead to accidents [2]. The methods to detect these road anomalies on expansive road networks generate unbearable costs for the third world countries. Developed countries use specialized vehicles with sensors to detect these anomalies. These vehicles are expensive and have high upkeep expenses. Therefore, we cannot rely on these methods for regular and timely inspections. The National Highway Authority (NHA) delays its surveys by 2-4 years due to insufficient funds. During this time, even a minor crack can develop into a large, irreparable pothole, significantly increasing maintenance costs [3].

1.1 Problem Statement

Imagine driving down the road at 100 miles per hour when you encounter an unexpected bump that causes you to slam on the brakes and you lose control of your vehicle. We've all faced such situations. In Pakistan, poor road conditions lead to vehicle damage, financial loss, and, tragically, even fatal accidents. Businesses also suffer due to poor infrastructure. Rough roads can damage goods during transport. NHA holds a survey after every 3- 5 years of all roads, this is a very time consuming method. Although a new solution is necessary, it's worth mentioning that the sector already has a couple of solutions in place. Road inspections today rely very much on manual checking, where workers go out, inspect the roads, and report issues. This method is incredibly slow and can't keep up with the vast network of roads that bear heavy traffic. Alternatively, in developed countries specialized vehicles are used that are effective. However, they are very expensive to operate and maintain. NHA used these cars for the surveys but due to the high costs of maintenance made them impractical. Additionally, both methods require a lot of manpower, making them inefficient , especially for a country like Pakistan with its extensive road networks that are poorly maintained. To counter the inefficiencies of these systems we're offering a system which relies on sensors that are commonly found in smart phones to report road conditions [1].

1.2 Scope

The scope of the project involves the development of both mobile and web applications designed for easy user interaction. The mobile application will be responsible for data collection, leveraging smartphone sensors. The smartphone sensors used for data collection will be the accelerometer, gyroscope, magnetometer, and GPS. Both the mobile and web applications will display maps showcasing road surface and elevation conditions, as well as dashboards that present statistics and insights into the collected data. The mobile app will also visualize real-time sensor values as line graphs. The data collection will be limited to the main roads of Islamabad and Rawalpindi, ensuring coverage across a variety of road settings. The project's reach extends to individual commuters, government bodies, logistics and transport companies. It offers benefits in road safety, operational efficiency, and cost reduction.

The scope of this project is limited by the following constraints:

- The application relies on internet connectivity to fetch and update data. Poor or no internet connection may result in a poor user experience.
- Our system relies on the static placement of mobile phones inside vehicles.
- As our system depends on crowd-sensing data, malicious actors may be able to send false data.

1.3 Modules

The project is divided into multiple modules, each addressing a distinct aspect of the system. Below is an overview of the primary modules:

1.3.1 Mobile App Module

The mobile app will be used to collect data from smartphone sensors. It will also show a dashboard of the data collected and the map of road surface conditions and elevation.

1. Data Collection Through Smartphone Sensors (Accelerometer, Gyroscope, Magnetometer)
2. User Authentication/User Management
3. Data Aggregation to a Central Database
4. Dashboard of Data Collected
5. Real-time Line Graphs of Sensor Values
6. Road Surface Condition Map
7. Road Elevation Map

1.3.2 Web App Module

The web app will be used to display a dashboard showcasing the statistics of the data collected along with a map highlighting road surface conditions and elevation, easily accessible through any web browser.

1. Visualization of Road Surface Conditions on a Map
2. Elevation Mapping of the Road Networks
3. User Login and Management Functionalities
4. Dashboard Showing Insights from Collected Data

1.3.3 Cloud-Based Inference Module

Initially, a model will be trained using the data collected from the mobile app. Once the model is trained, it will be deployed to the cloud, where it will continuously perform inference on incoming data to detect road anomalies such as potholes, bumps, and rough surfaces.

1. Model Training
2. Cloud Deployment
3. Scheduled Batch Inference

1.4 User Classes and Characteristics

User class	Description
Government Authorities	Government authorities like the RDA, CDA, and NHA are responsible for monitoring road conditions to prioritize maintenance and repairs, ensuring public safety by keeping roads in good condition. These authorities are likely to use the application for large-scale data collection on road conditions. Additionally, they can plan and allocate budgets more efficiently based on the collected data. They could also integrate the system with public works or urban planning systems.
Freight Companies	The main concern is maintaining the condition of the fleet by avoiding or minimizing damage caused by potholes. Another challenge is reducing vehicle repair costs and downtime resulting from poor road conditions. They also need a mechanism to optimize routes for faster and safer deliveries. The system can be integrated into their fleet management system for enabling proactive decision-making. This can help increase vehicle efficiency and reduce maintenance costs. Additionally, they can operate heavy vehicles more safely and efficiently, improving fuel consumption.
Everyday Drivers	The major concern for everyday drivers is avoiding potholes to prevent vehicle damage, reduce the risk of accidents, and avoid unexpected repair costs. They can use this application to receive information about damaged areas. Drivers can also provide data on road conditions, enhancing their convenience, ease of use, and clear communication of road hazards.
Emergency Responders	The utmost priority is quick access to emergency sites without vehicle damage. They also need to avoid delays caused by poor road conditions. This system will provide reliable data to help plan the fastest and safest routes, which is crucial when operating under highly sensitive conditions, such as transporting a patient. For this, they require a system that is reliable and constantly updated.
Tourists	When traveling on unfamiliar roads, they want to navigate safely and efficiently, avoiding bad roads to ensure a smooth trip. They seek confidence in road conditions, especially when driving a rented vehicle. This application can help them navigate new roads while avoiding any discomfort. They are also concerned with easy access to road information, especially when they are not familiar with the local area.
Data Contributors	People use the app solely to contribute data, voluntarily sharing sensor data from their mobile phones to help improve public knowledge about road conditions.

Table 1.1: User Classes and Their Descriptions

Chapter 2

Project Requirements

This chapter outlines the project's functional and non-functional requirements, including use cases to clarify system functionality.

2.1 Use Cases

2.1.1 Use Case Diagram

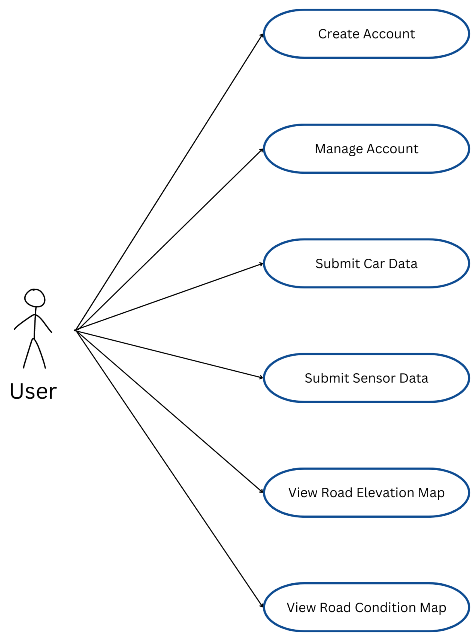

Figure 2.1: Use Case Diagram for the Application

2.1.2 Expanded Use Cases

The expanded use cases for the application are detailed below:

2.1.2.1 UC-01: User Authentication

Use Case ID	UC-01
Name	User Authentication
Scope	Login, Registration, and Authentication
Level	User goal
Primary Actor	Commuter, Freight Companies
Stakeholders and Interests	Commuter, Freight Companies: Want secure and seamless access to their accounts and personalized settings.
Pre-condition	User must provide valid credentials or register a new account.
Success Guarantee	User successfully logs in or registers, accessing the app features.
Main Success Scenario	1. User opens the app and goes to the login page. 2. User enters credentials or chooses to register a new account. 3. The system verifies the user credentials.
Extensions	2a: If the user clicks "Forgot Password," the system prompts for their email, sends a reset link, and allows them to reset their new password. 3a: System notifies the user of invalid credentials and prompts them to re-enter the correct credentials
Special Requirements	The system must handle secure encryption for password storage

Table 2.1: UC-01: User Authentication

2.1.2.2 UC-02: Crowdsource Road Data

Use Case ID	UC-02
Name	Crowdsource Road Data
Scope	Collecting road condition data from mobile sensors
Level	User goal
Primary Actor	Commuter, Freight Companies
Stakeholders and Interests	Commuter/Freight Companies: Interested in contributing data to improve road condition accuracy.
	Government Authorities/Emergency Responders: Rely on high-quality, reliable crowd-sourced data for decision-making.
Pre-condition	User has consented to share road condition data; app has permission to use sensors; GPS is enabled.
Success Guarantee	The app successfully collects and uploads accurate road condition data from the user's smartphone sensors.
Main Success Scenario	1. User logs in and starts driving. 2. The app uses smartphone sensors (accelerometer, gyroscope, magnetometer etc.) to collect data. 3. Data is uploaded to a central database.
Extensions	2a: If internet connection fails, data is cached for future upload.
Special Requirements	Ability to cache data offline for later upload.

Table 2.2: UC-02: Crowdsource Road Data

2.1.2.3 UC-03: Collect Car Data

Use Case ID	UC-03
Name	Collect Car Data
Scope	Gathering vehicle information from users
Level	User goal
Primary Actor	User (Commuter, Freight Companies)
Stakeholders and Interests	App Developers: Need data for model training.
Pre-condition	User must be logged in and opt to share data.
Success Guarantee	The app successfully collects and stores vehicle data.
Main Success Scenario	1. User logs in. 2. User selects the option to enter car details. 3. User inputs make, model, year, and condition of the car. 4. The app confirms that data has been saved successfully.
Extensions	2a: If the user cancels, data entry is terminated.
Special Requirements	User interface for data input.

Table 2.3: UC-03: Collect Car Data

2.1.2.4 UC-04: Road Elevation Map

Use Case ID	UC-04
Name	Road Elevation Map
Scope	Visualizing road elevation data
Level	User goal
Primary Actor	User (Commuter, Government Authorities, Freight Companies, Emergency Services)
Stakeholders and Interests	Commuters: Need elevation data for route planning Government Authorities: Require elevation data to monitor infrastructure and plan road maintenance or development projects. Freight Companies: Use elevation data to optimize delivery routes and do not go through areas that could increase vehicle wear and tear or fuel consumption. Emergency Services: Rely on elevation data to efficiently and safely navigate during emergency response, particularly in regions containing mountains.
Pre-condition	User must be logged in, and location data must be enabled.
Success Guarantee	The app successfully displays an accurate elevation map of the selected route.
Main Success Scenario	1. User logs in. 2. User selects the option to view the road elevation map. 3. User selects or inputs a route to map. 4. The app displays the elevation map for the chosen route.
Extensions	2a: If the user cancels, the map rendering is terminated.
Special Requirements	Accurate elevation data, user interface to view the map.

Table 2.4: UC-04: Road Elevation Map

2.1.2.5 UC-05: Road Condition Map

Use Case ID	UC-05
Name	Road Condition Map
Scope	Displaying road conditions for safe navigation
Level	User goal
Primary Actor	User (Commuter, Government Authorities, Freight Companies, Emergency Services)
Stakeholders and Interests	Commuters: Need up-to-date road condition information to plan safer journeys. Government Authorities: Utilize data to assess road maintenance needs and ensure road safety. Freight Companies: Need information on road conditions to optimize delivery routes and reduce delays. Emergency Services: Rely on accurate data to ensure safe and efficient access to affected areas.
Pre-condition	User must be logged in, and location services must be enabled.
Success Guarantee	The app successfully displays up-to-date road condition data on the map.
Main Success Scenario	1. User logs in. 2. User selects the option to view the road condition map. 3. User selects or selects a specific area to view. 4. The app displays current road conditions.
Extensions	2a: If the user cancels, data loading is terminated. 3a: If no data is available, the app just displays a simple map.
Special Requirements	A user-friendly interface that shows a map of road networks.

Table 2.5: UC-05: Road Condition Map

2.2 Functional Requirements

This section describes the functional requirements of the system. The system is divided into three main modules, each with specific features and functional requirements.

2.2.1 Mobile App Module

The mobile app will be used to collect data from smartphone sensors and visualize it through graphs. The app will also provide a dashboard of data collected and maps of road surface conditions and elevation.

2.2.1.1 FR1-001: User Authentication and Management

Field	**Information**
Title	User Authentication and Management
ID	FR1-001
Version	1.0
Description	The system shall allow users to create accounts, log in, and manage their profiles through the mobile app.
Priority	High
Rationale	Ensures that users can securely access and manage their data in the app.
Cross-Reference	UC-01: User Authentication (2.1)

Table 2.6: FR1-001: User Authentication and Management

2.2.1.2 FR1-002: Data Aggregation to Database

Field	**Information**
Title	Data Aggregation to Database
ID	FR1-002
Version	1.0
Description	The system shall aggregate the sensor data collected from multiple users and store it in a central database.
Priority	High
Rationale	Will be used to predict road elevation and surface condition.
Cross-Reference	Refer to Section 3.3.2

Table 2.7: FR1-002: Data Aggregation to Database

2.2.1.3 FR1-003: Dashboard of Data Collected

Field	**Information**
Title	Dashboard of Data Collected
ID	FR1-003
Version	1.0
Description	The system shall provide a dashboard in the mobile app that shows an overview of the data collected by the sensors.
Priority	Medium
Rationale	Provides users with an overview and insights into the data that has been collected.
Cross-Reference	Refer to Section 1.3.1

Table 2.8: FR1-003: Dashboard of Data Collected

2.2.1.4 FR1-004: Real-time Line Graphs of Sensor Values

Field	**Information**
Title	Real-time Line Graphs of Sensor Values
ID	FR1-004
Version	1.0
Description	The system shall display real-time graphs in the mobile app that visualize the data collected from the accelerometer, gyroscope, and magnetometer sensors.
Priority	Medium
Rationale	Helps users visually understand the data captured by sensors in real time.
Cross-Reference	Refer to Section 1.3.1

Table 2.9: FR1-004: Real-time Graphs of Sensor Values

2.2.1.5 FR1-005: Road Surface Condition Map

Field	**Information**
Title	Road Surface Condition Map
ID	FR1-005
Version	1.0
Description	The system shall display a map within the mobile app that highlights road surface conditions based on the data collected from users.
Priority	High
Rationale	Provides users with detailed road surface condition insights.
Cross-Reference	Refer to Section 1.3.1, UC-05: Road Condition Map (2.5)

Table 2.10: FR1-005: Road Surface Condition Map

2.2.1.6 FR1-006: Road Elevation Map

Field	**Information**
Title	Road Elevation Map
ID	FR1-006
Version	1.0
Description	The system shall display a road elevation map in the mobile app.
Priority	High
Rationale	Provides valuable elevation information for users planning trips to mountainous regions.
Cross-Reference	Refer to Section 1.3.1, UC-04: Road Elevation Map (2.4)

Table 2.11: FR1-006: Road Elevation Map

2.2.2 Web App Module

The web app will allow users to view road surface conditions and elevation map and enable them to login and manage their account.

2.2.2.1 FR2-001: User Authentication and Management (Web)

Field	**Information**
Title	User Authentication and Management
ID	FR2-001
Version	1.0
Description	The web app shall shall provide users with functionality that enables them to create accounts, login, and manage their profiles.
Priority	High
Rationale	The app ensures secure access and management of user data.
Cross-Reference	Refer to Section 1.3.2, UC-01: User Authentication (2.1)

Table 2.12: FR2-001: User Account Management (Web)

2.2.2.2 FR2-002: Visualization of Road Surface Conditions on a Map (Web)

Field	**Information**
Title	Visualization of Road Surface Conditions on a Map
ID	FR2-002
Version	1.0
Description	The system shall display a map on the web app that visualizes the road surface conditions, based on the data collected from users' mobile devices.
Priority	High
Rationale	Provides users with visual insights into road surface conditions.
Cross-Reference	Refer to Section 1.3.2, UC-05: Road Condition Map (2.5)

Table 2.13: FR2-002: Road Surface Conditions (Web)

2.2.2.3 FR2-003: Elevation Mapping of the Road Networks (Web)

Field	**Information**
Title	Elevation Mapping of the Road Networks
ID	FR2-003
Version	1.0
Description	The system shall display a map in the web app that shows the elevation of roads using the data from users' mobile devices.
Priority	High
Rationale	Provides users with a visualization of road elevation data, aiding in navigation.
Cross-Reference	Refer to Section 1.3.2, UC-04: Road Elevation Map (2.4)

Table 2.14: FR2-003: Elevation Map (Web)

2.2.2.4 FR2-004: Data Dashboard (Web)

Field	**Information**
Title	Dashboard Showing Insights from Collected Data
ID	FR2-004
Version	1.0
Description	The system shall provide a dashboard in the web app that offers insights and statistics from the aggregated sensor data.
Priority	High
Rationale	Allows users to see an overview of the collected data.
Cross-Reference	Refer to Section 1.3.2

Table 2.15: FR2-004: Data Dashboard (Web)

2.2.3 Cloud-Based Inference Module

This module will handle the cloud-based machine learning aspects of the project, including model training, deployment, and batch inference.

2.2.3.1 FR3-001: Model Training

Field	Information
Title	Model Training
ID	FR3-001
Version	1.0
Description	The system shall train a machine learning model using the data collected from users' smartphones to detect anomalies in road surfaces (e.g., potholes, bumps, rough surfaces).
Priority	High
Rationale	Detecting road anomalies improves road safety and reduces maintenance costs.
Cross-Reference	Data Collection 2.7 (FR1-002), Refer to Section 1.3.3

Table 2.16: FR3-001: Model Training

2.2.3.2 FR3-002:Cloud Deployment

Field	Information
Title	Cloud Deployment
ID	FR3-002
Version	1.0
Description	The system shall deploy the trained machine learning model to a cloud environment, enabling it to perform inferences on data.
Priority	High
Rationale	Cloud deployment allows scalable and accessible model inferences.
Cross-Reference	Refer to Section 1.3.3

Table 2.17: FR3-001: Model Deployment

2.2.3.3 FR3-003:Scheduled Batch Inference

Field	**Information**
Title	Scheduled Batch Inference
ID	FR3-003
Version	1.0
Description	The system shall schedule regular batch inferences using the trained model to analyze incoming data for road anomalies, which will then be visualized on the road condition map.
Priority	High
Rationale	Regular batch processing ensures that the road condition data is up-to-date.
Cross-Reference	Refer to Section 1.3.3

Table 2.18: FR3-003: Scheduled Batch Inference

2.3 Non-Functional Requirements

This section specifies the nonfunctional requirements of the system.

2.3.1 NFR-ACC-001: Accessibility

Field	**Information**
ID	NFR-ACC-001
Title	Accessibility
Description	The application shall be accessible to a diverse range of users, with a responsive GUI across various devices such as computers, tablets, and mobile phones. The interface must be simple and intuitive to ensure ease of use for users with minimal technical expertise.
Rationale	Ensures that the system can be used by a wider range of users, increasing the potential user base.
Validation	Test the application across various devices and collect feedback from users with varying technical skills.

Table 2.19: NFR-ACC-001: Accessibility

2.3.2 NFR-ADP-001: Adaptability

Field	**Information**
ID	NFR-ADP-001
Title	Adaptability
Description	The system shall employ techniques to improve its performance over time by adapting to new data. It will aim to reduce prediction errors by 10% annually and remain accurate despite changes in patterns and trends.
Rationale	Continuous improvement in prediction accuracy ensures the system remains relevant and effective.
Validation	Periodic evaluations of system accuracy and error rates based on the latest data should be conducted.

Table 2.20: NFR-ADP-001: Adaptability

2.3.3 NFR-ACCY-001: Accuracy

Field	**Information**
ID	NFR-ACCY-001
Title	Accuracy
Description	The system's AI algorithm shall achieve an accuracy of at least 90% in predicting road surface conditions.
Rationale	High accuracy ensures reliability in the system's predictions, increasing user trust.
Validation	Test the prediction accuracy using a test set and validate with real-world data.

Table 2.21: NFR-ACCY-001: Accuracy

2.3.4 NFR-AVL-001: Availability

Field	Information
ID	NFR-AVL-001
Title	Availability
Description	The system shall maintain an uptime of 98% or above, ensuring users can reliably access road conditions at any time. Downtime should be minimal and maintenance shall be scheduled during periods of low usage.
Rationale	High availability is critical for user satisfaction and consistent access to the system.
Validation	Monitor system uptime and ensure scheduled maintenance does not affect user experience.

Table 2.22: NFR-AVL-001: Availability

2.3.5 NFR-RT-001: Response Time

Field	Information
ID	NFR-RT-001
Title	Response Time
Description	The system's database response time shall be less than 1 second for data retrieval, ensuring smooth and efficient user experience.
Rationale	Quick response time improves user interaction and reduces latency in accessing information.
Validation	Regular performance testing should be conducted to ensure the response time meets the specified threshold.

Table 2.23: NFR-RT-001: Response Time

2.3.6 NFR-SEC-001: Data Security

Field	**Information**
ID	NFR-SEC-001
Title	Data Security
Description	The system shall encrypt user data both in transit and at rest, using industry-standard security protocols to safeguard it from unauthorized access.
Rationale	Ensures the confidentiality, integrity, and security of sensitive user information.
Validation	Perform regular security audits and implement encryption protocols for data protection.

Table 2.24: NFR-SEC-001: Data Security

2.4 Domain Model

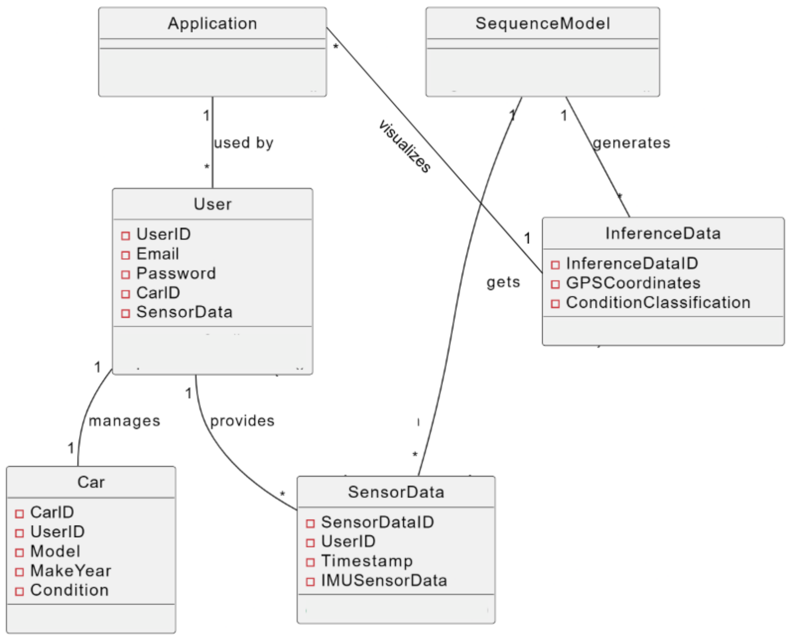

Figure 2.2: Domain Model Diagram of the System

Chapter 3

System Overview

This chapter provides a comprehensive overview of the design models utilized in our project. We will discuss the architectural design, detailing the modular structure of our program and the relationships between the system's components. The focus will be on understanding how the system is decomposed into subsystems and how these subsystems work together to achieve the desired functionality. We'll also go over the flow of data in our system and how our data is structured and stored inside databases.

3.1 Architectural Design

3.1.1 Communication Flow

The flow of data between the layers is as follows:

1. The Data Access Layer takes the data from the Hardware Layer and passes it to the Data Storage Layer.

2. The data is then sent to the Business Logic Layer, which processes it.

3. The processed results are subsequently provided to the Presentation Layer, which displays them in a user-friendly format.

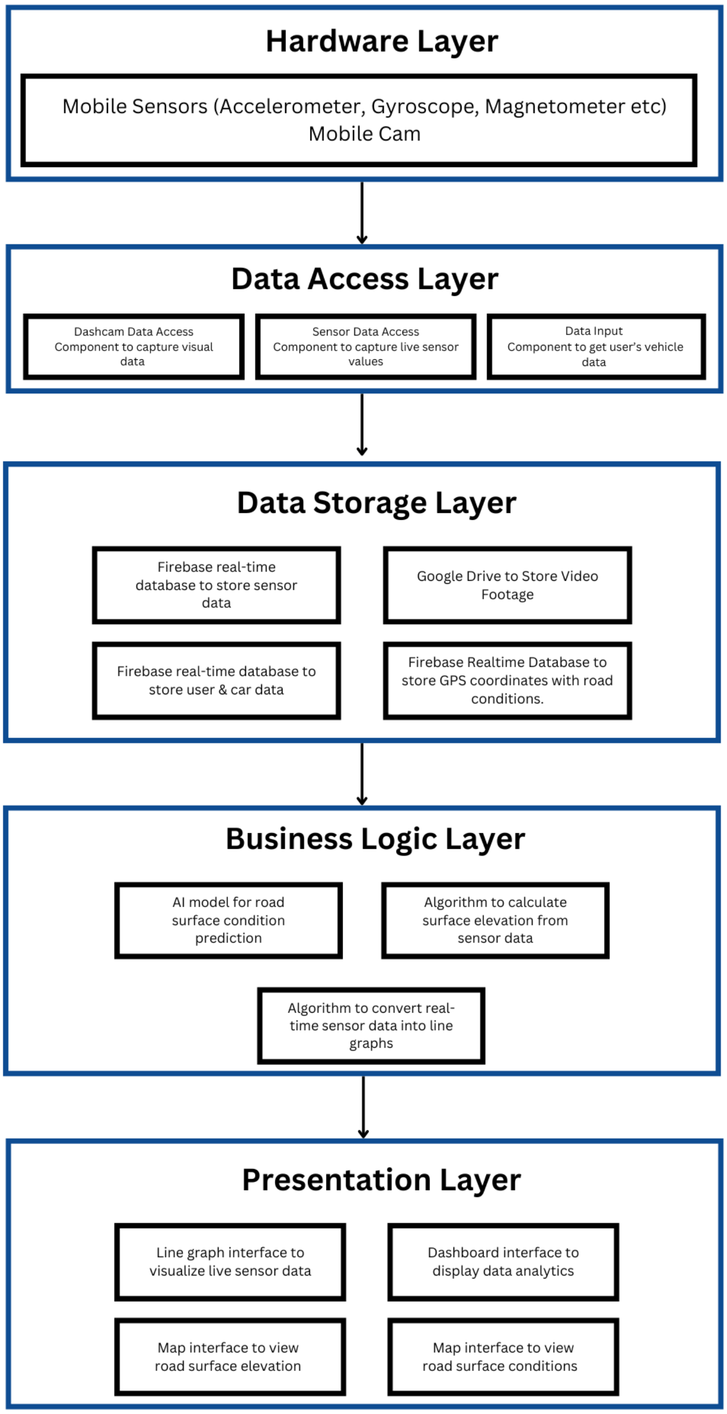

Figure 3.1: System Architecture Design

3.1.2 Diagram Explanation

3.1.2.1 1. Hardware Layer

The mobile sensors and mobile camera are part of this layer. This layer has the data needed to train the model to determine road conditions.

3.1.2.2 2. Data Access Layer

The Data Access Layer is built upon the Hardware Layer, serving as the interface that connects software with hardware components. It contains code that facilitates communication and data transfer between mobile sensors, cameras, and the application.

3.1.2.3 3. Data Storage Layer

Our Firebase databases are part of this layer. These databases securely store our data in a structured manner for efficient retrieval.

3.1.2.4 4. Business Logic Layer

The algorithms and the AI model are part of this layer. This layer is the brain of our system, processing data and implementing functionalities.

3.1.2.5 5. Presentation Layer

Components of the user interface, such as the Dashboard, Line Graphs, and Maps, are part of this layer. Their purpose is to display and visualize data for the user.

3.2 Design Models

3.2.1 Activity Diagram

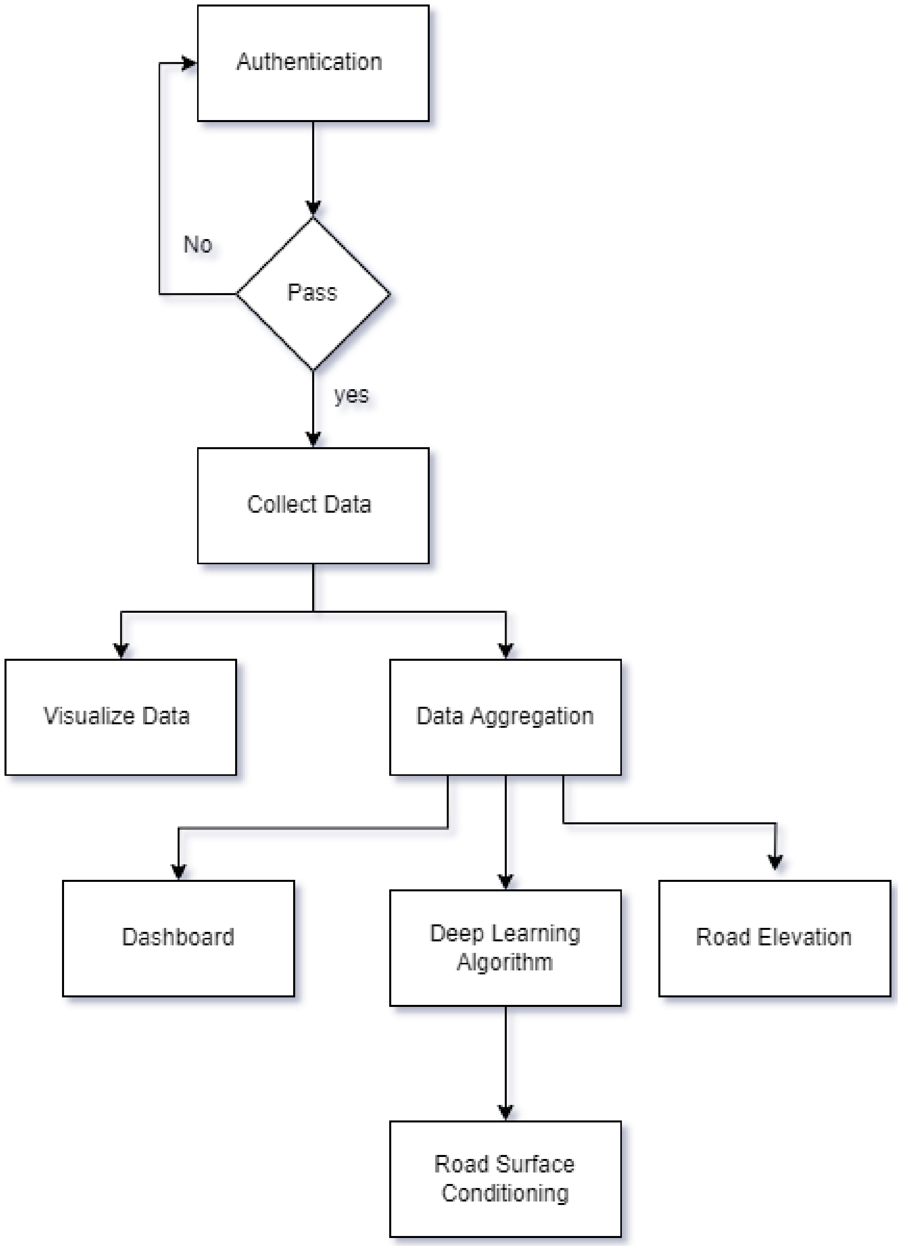

Figure 3.2: Activity Diagram

3.2.1.1 Diagram Explanation:

The activity diagram illustrates the process flow of a system that authenticates users, collects, aggregates data, and provides outputs related to road surface conditions and elevation. The process begins with user authentication, where access is granted if the user inputs valid credentials. Upon successful authentication, data collection is initiated on user's consent. The collected data can then be visualized through graphs. The data is also aggregated to a central database. Insights about the aggregated data can be seen on the data dashboard. Data aggregation also feeds a deep learning algorithm and computes road elevation. The results of the deep learning algorithm provide information on road surface conditions, which is the primary output of the system.

3.2.2 Data Flow Diagram

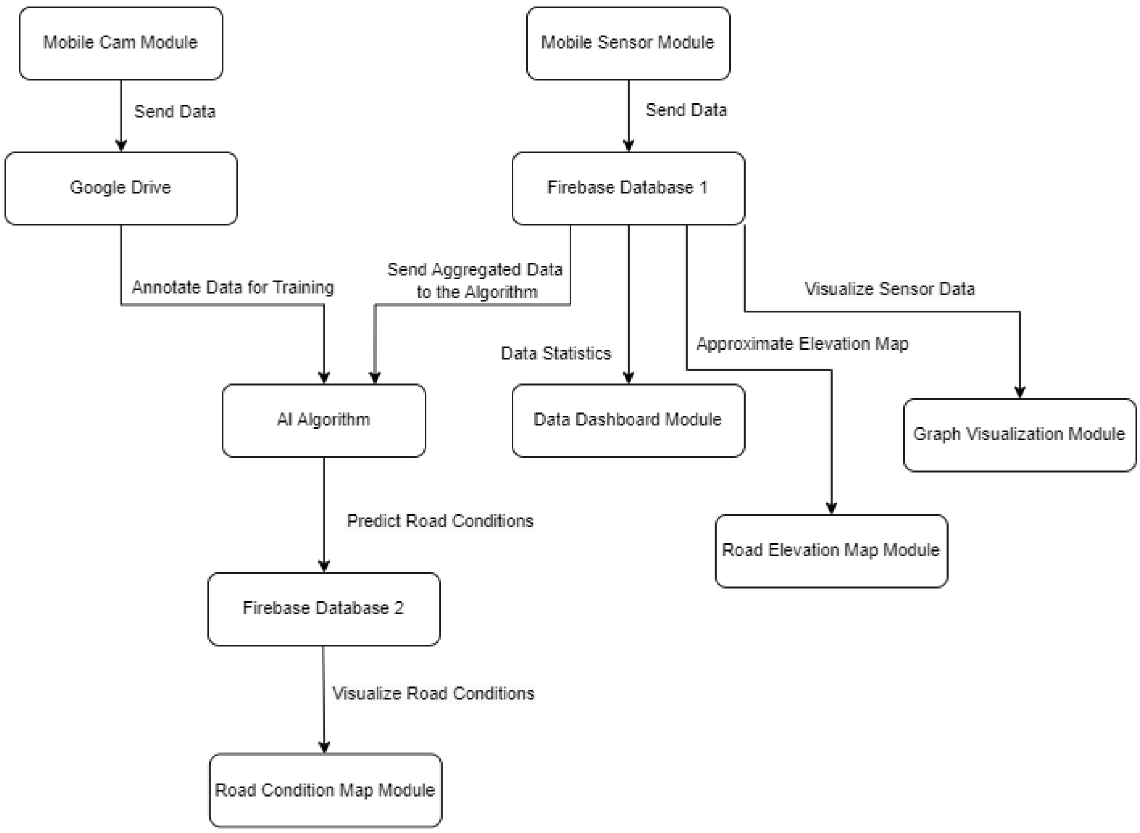

Figure 3.3: Data Flow Diagram

3.2.2.1 Diagram Explanation:

The data flow diagram demonstrates the interaction between various system modules to predict road conditions and visualize road surface data. The process starts with two data sources: the Mobile Cam Module and the Mobile Sensor Module. The Mobile Cam Module sends video data to Google Drive, where it is annotated for training purposes.

Meanwhile, the Mobile Sensor Module sends sensor data to Firebase Database 1, which aggregates the data and provides it to the AI algorithm for training and prediction.

The aggregated data from Firebase Database 1 is sent to the AI algorithm, which predicts road conditions. These predictions are stored in Firebase Database 2 and subsequently visualized through the Road Condition Map Module.

In parallel, Firebase Database 1 also sends data to the Data Dashboard Module for statistical analysis and to the Graph Visualization Module as well as the road elevation maps module.

3.2.3 System-level Sequence Diagram

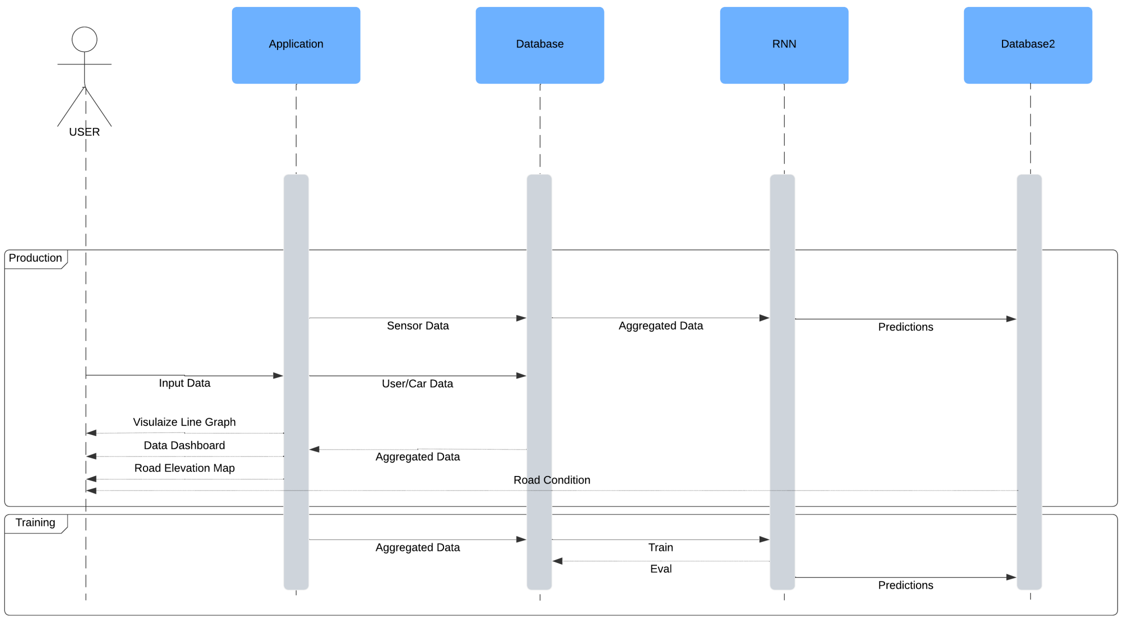

Figure 3.4: System-level Sequence Diagram

3.2.3.1 Diagram Explanation:

The diagram showcases the sequence model of the RoadInsight system. The system is divided into two main phases: Production and Training.

Production Phase:

- The user provides data through their smartphones, which is aggregated by the system and stored in a database.
- The data collected includes sensor data and user/car data.
- Aggregated data is forwarded to the AI Model, which makes predictions based on this data.
- Aggregated data is also used to show data dashboards and road elevation maps to the user.
- Predictions about road conditions are then stored in a second database and made available through map visualizations for the user.

Training Phase:

- In the training phase, aggregated data from the database is used to train the AI model.
- The model is evaluated, and the predictions generated during training are used to improve the system's performance in the production phase.

The flow of system sequences, shows how user input is processed, and presented both in real-time and during the model's training cycle.

3.2.4 Use Case Sequence Diagrams

3.2.5 UC-01: User Authentication

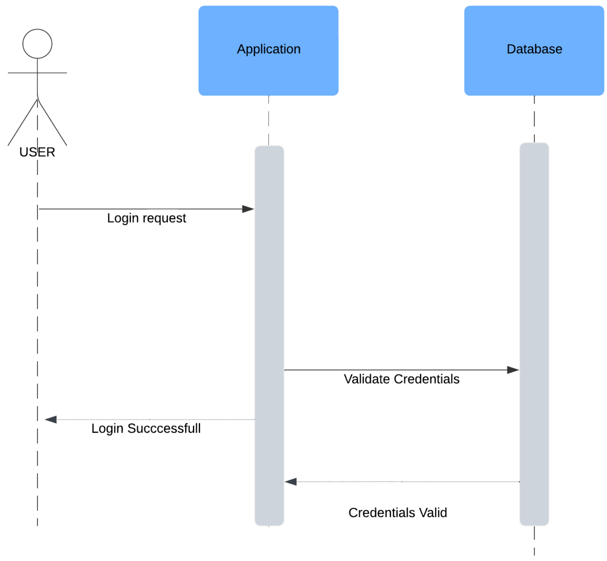

Figure 3.5: Sequence Diagram: UC-01

3.2.6 UC-02: Crowdsource Road Data

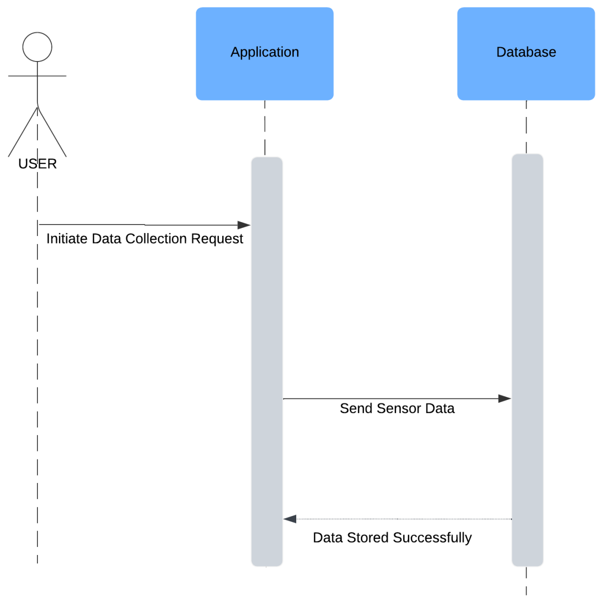

Figure 3.6: Sequence Diagram: UC-02

3.2.7 UC-03: Collect Car Data

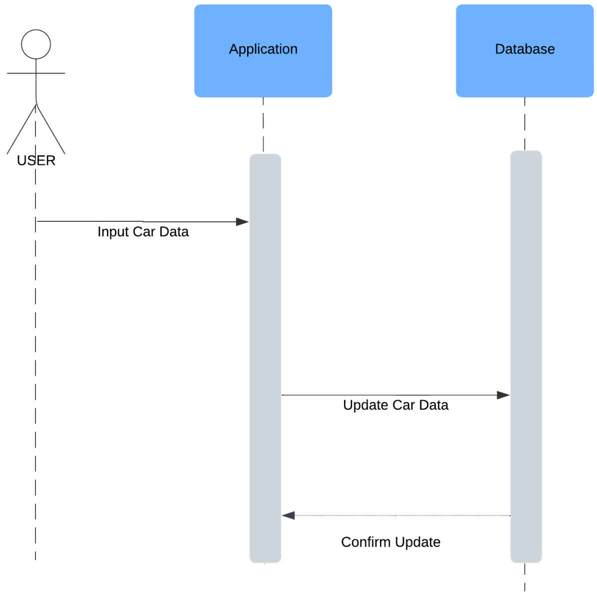

Figure 3.7: Sequence Diagram: UC-02

3.2.8 UC-04: Road Elevation Map

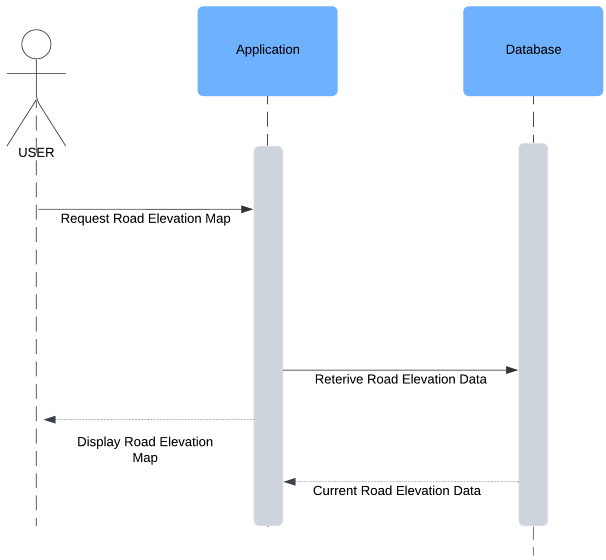

Figure 3.8: Sequence Diagram: UC-04

3.2.9 UC-05: Road Condition Map

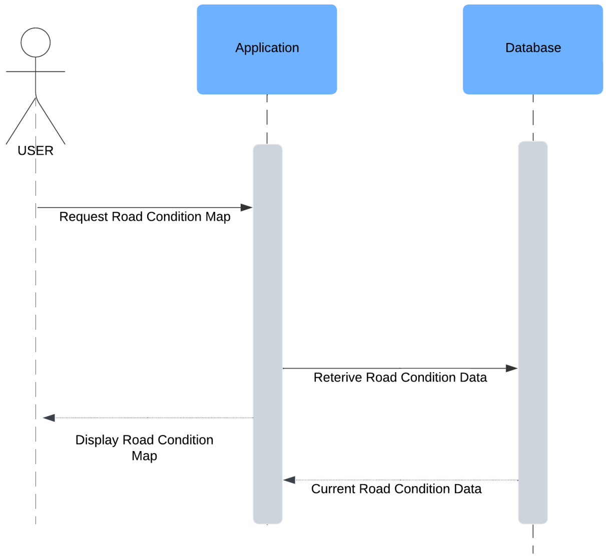

Figure 3.9: Sequence Diagram: UC-05

3.3 Data Design

Firebase serves as the primary data storage solution for our application. The information domain of the application is structured around key data entities that reflect the core functionalities of the system. These entities are transformed into data structures that are efficiently stored, processed, and organized within our Firebase databases.

3.3.1 Major Data Entities

- **User Data**
  - The application gets basic user information, including Email, and Password from the user's themselves. This data is required for user account creation and authentication processes. It is stored in a real-time NoSQL Firebase database, allowing for secure management of user credentials and efficient retrieval for authentication operations.
- **Car Data**
  - Each User can manage information about their vehicle, which includes attributes such as Model, Make Year, and Vehicle Condition. This data is also

stored in the same Firebase database as user data, creating a direct association between users and their vehicles.

- **Sensor Data**
  - Sensor data is collected from mobile phone sensors statically placed inside vehicles. This data includes Timestamp, GPS coordinates, and data from mobile sensors such as Accelerometer, Gyroscope and Magnetometer. Each data point is stored in the dedicated database.
- **Predictions Data**
  - The application leverages an RNN model to make predictions based on the collected sensor data. The results of these predictions and their GPS coordinates are stored in a separate Firebase database. This structured organization allows for easy access and retrieval of prediction results for display on the application's user interface.
- **Video Data**
  - Video data is captured using mobile cameras and uploaded to Google Drive. This data is utilized exclusively in the initial phases for annotating the data for model training.

3.3.2 Data Flow and Processing

This application follows a systematic approach for logical flow of data and data processing. During account registration, user information is collected and processed. This information will later be used to facilitate user authentication and session management. Each user's car data is also taken upon account creation. Car data is linked to the respective user, enabling unified experience when managing vehicle-related information.

Sensor data is collected in real-time from users and stored in the designated Firebase database. This data undergoes scheduled batch processing, where an RNN model infers predictions about road conditions from it. The predictions generated by the model are stored in a separate database, ensuring that they can be efficiently accessed and presented to users as needed. Additionally, the captured video data is used during the initial phases to annotate sensor data for training the model.

3.3.3 Databases and Data Storage

The system utilizes two primary Firebase NoSQL databases and Google Drive to store data:

- **Firebase Database 1:** Stores User and Car Data, facilitating user management and vehicle information. It also contains Sensor Data collected from mobile phones, essential for predicting road conditions.
- **Firebase Database 2:** Stores Predictions Data derived from the AI model, providing insights on road conditions based on sensor inputs.
- **Google Drive:** Stores Video Data captured from mobile cameras, used for training the model in the initial phases.

This structured data design ensures that the application can effectively manage, process, and display information relevant to users, and provide a unified experience while providing valuable insights into road conditions.

Bibliography

[1] A Alrajhi, K Roy, L Qingge, and J Kribs. Detection of road condition defects using multiple sensors and iot technology: A review. *IEEE Open Journal of Intelligent Transportation Systems*, 4:372–392, 2023.

[2] Correspondent. Damaged roads can cause fatal accidents, 2022. Accessed: 2024-09-13.

[3] Kris Jepson. North east pothole repairs backlog will cost £820m, says new survey, March 2024. Accessed: September 13, 2024.